G.4 Implementation-Oriented Comparison

This table explains what would be required to replace or extend the current predictor in the existing Flask application.

Algorithm	New data or state required	Parameters / tuning	Integration change	Validation needed	Recommended decision
Constant velocity and heading	Current latitude, longitude, speed, heading and vertical rate.	Maximum age and physical sanity limits.	Already available as the regression fallback.	Replay tests for straight, climbing and descending motion.	Retain as a safe fallback, not the primary predictor.
Sliding-window linear regression (USED)	Recent timestamped latitude, longitude and altitude samples.	Window size, minimum sample count and maximum prediction age.	Already implemented in the storage prediction path.	Measure horizontal and altitude error at 5-180 s gaps.	Keep as the explainable prototype baseline.
Kalman filter [10]	Position and velocity state plus uncertainty estimates.	Process noise, measurement noise and initial covariance.	Add a per-aircraft filter state and reset rules.	Test noisy data, missing updates and manoeuvre transitions.	Preferred next upgrade after quantitative baseline testing.
Polynomial regression	Recent samples with enough time spread for a stable curve fit.	Polynomial order, window length and strict extrapolation horizon.	Replace the linear fit while retaining current history queries.	Check overshoot, oscillation and sensitivity to outliers.	Use only as a controlled experiment.
Random forest / boosting [11]	Engineered features and labelled future-position targets.	Tree depth, ensemble size and feature design.	Add offline training, model loading and version control.	Use held-out flights and prevent route leakage between splits.	Defer until a representative labelled dataset exists.
LSTM / GRU [12]-[14]	Long normalized trajectory sequences and future target sequences.	Architecture, sequence length, optimizer and regularization.	Add training pipeline, model runtime and hardware planning.	Use route-separated validation, uncertainty checks and drift monitoring.	Future research only; excessive for the current dataset.

Selection conclusion: retain linear regression for transparency; evaluate Kalman filtering next using measured gap-prediction error.

Table 16: Implementation-Oriented Algorithm Comparison